

More SQL

MBUA 512

Class 4 – More SQL

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

- SELECT calculations, ORDER BY (DESC), LIMIT n, named columns, DISTINCT
- WHERE conditions
- GROUP BY, HAVING, and aggregates (min, max, avg, count, sum)

How to write SQL queries & tips

Based on slides by Professor Markus Luczak-Roesch.

Recap: SQL commands

- `CREATE TABLE` and `ALTER TABLE` — define the structure of a table (its *schema*)
 - `INSERT INTO` , `DELETE FROM` , `UPDATE` — change the data in the table
 - `SELECT` — query data
- (there are a lot more than these)

Important: make sure you select `MySQL` (not `Trino`) and your own workspace
(`user_[username]`)

SELECT named columns

Can use `AS name` to give a column a name

```
SELECT concat(firstName, ' ', lastName)
SELECT concat(firstName, ' ', lastName) AS Name
```

	CONCAT(firstName, ' ', lastName) ⌵
1	Brice Taffurelli
2	Delmar Arrigo
3	Edy Jeyness

	Name ⌵
1	Brice Taffurelli
2	Delmar Arrigo
3	Edy Jeyness

Use quotes if the name has spaces, e.g. `AS 'Pilot name'`

SELECT DISTINCT

- Just add `DISTINCT` to remove duplicates *rows* from the result
- Example: select all pilots who have scheduled flights

```
select pilotName from flight, pilot where flight.pilotID=pilot.pilotID  
select DISTINCT pilotName from flight, pilot where flight.pilotID=pilot.pilotID
```

	pilotName
1	Pat Smith
2	Yi Zhang
3	Jim Belich
4	Jim Belich
5	Jim Belich
6	Jim Belich

	pilotName
1	Pat Smith
2	Yi Zhang
3	Jim Belich

SELECT calculations, ORDER BY, LIMIT

SELECT can include calculations:

```
SELECT concat(firstName, ' ', lastName) AS Name,  
       floor(datediff(current_date, birthdate)/365) AS Age  
FROM people WHERE birthdate IS NOT NULL  
ORDER BY floor(datediff(current_date, birthdate)/365) DESC,  
       astName  
LIMIT 10;
```

Explanation: `concat` joins strings, `current_date` is the current date, `datediff` gives the difference between two dates in days, `floor` rounds a number down. `EXTRACT(MONTH from d)` gets the month part of a date (not shown).

	Name	Age
1	Bernette Blance	98
2	Dion Lantiff	98
3	Pincas Sapwell	98
4	Sancho Puddephatt	96
5	Harman Ramey	95
6	Gun Beardsley	94
7	Tyrus Forte	94
8	Delmar Arrigo	89
9	Price Guerin	87
10	Edlin Lenard	87

WHERE conditions

All sorts of things you can test as conditions in the WHERE part.

- IS NULL - is something NULL? (e.g. `email IS NULL`) - also IS NOT NULL.
- IN: `month IN (1,2,3)` - shorthand for `month=1 OR month=2 OR month=3`.
- LIKE: is a string like a pattern, where `%` means "any string of characters" and `_` means any single character (e.g. `email LIKE '%@%.%'`).
- numerical conditions (`<` `>` `=` `>=` `!=`)
- date conditions (`<` `=` between): `d BETWEEN d1 AND d2`
- logical operators (AND, OR, NOT)
- (not) EXISTS (next slide)

EXISTS

- A WHERE condition is applied to each row on its own. But sometimes we need to check *other* rows when deciding whether to include a given row
- Example: find all people who share a birthday: the condition for a row is whether someone *else* (another row) has the same birthday.
- An EXISTS condition allows us to check a condition for a given row against more than just that row.

```
SELECT personID, firstName, lastName, birthdate FROM people p
WHERE EXISTS (select * from people p2 where p.birthdate=p2.birthdate AND p.personID != p2.personID);
```

Note use of names (p1 , p2) to refer to row in the subquery.

UNION, INTERSECT

- These combine the results of queries.
- UNION: include all rows from both queries
- INTERSECT: include only rows that are in both results. Example: all pilots who fly both a Boeing and an Airbus (can't do this with a WHERE!)

```
(SELECT pilotName from flight, pilot, aircraft
WHERE flight.pilotID=pilot.pilotID AND flight.planeModel=aircraft.planeModel
AND planeManufacturer='Airbus')
INTERSECT
(SELECT pilotName from flight, pilot, aircraft
WHERE flight.pilotID=pilot.pilotID AND flight.planeModel=aircraft.planeModel
AND planeManufacturer='Boeing');
```

GROUP BY, HAVING, aggregates

- *Aggregates* combine rows (min, max, avg, count, sum), by default all rows
- *GROUP BY* specifies which rows to combine
 - When grouping by something (e.g. day) include it in the output
 - WARNING: including a non-aggregate in the output other than what's used for grouping gives an arbitrary element
- *HAVING* filters out certain *groups* (WHERE filters out *rows*)

```
SELECT min(time) FROM flightinfo; -- earliest flight
```

```
SELECT day, min(time), max(time), count(time)
```

```
FROM flightinfo, flight
```

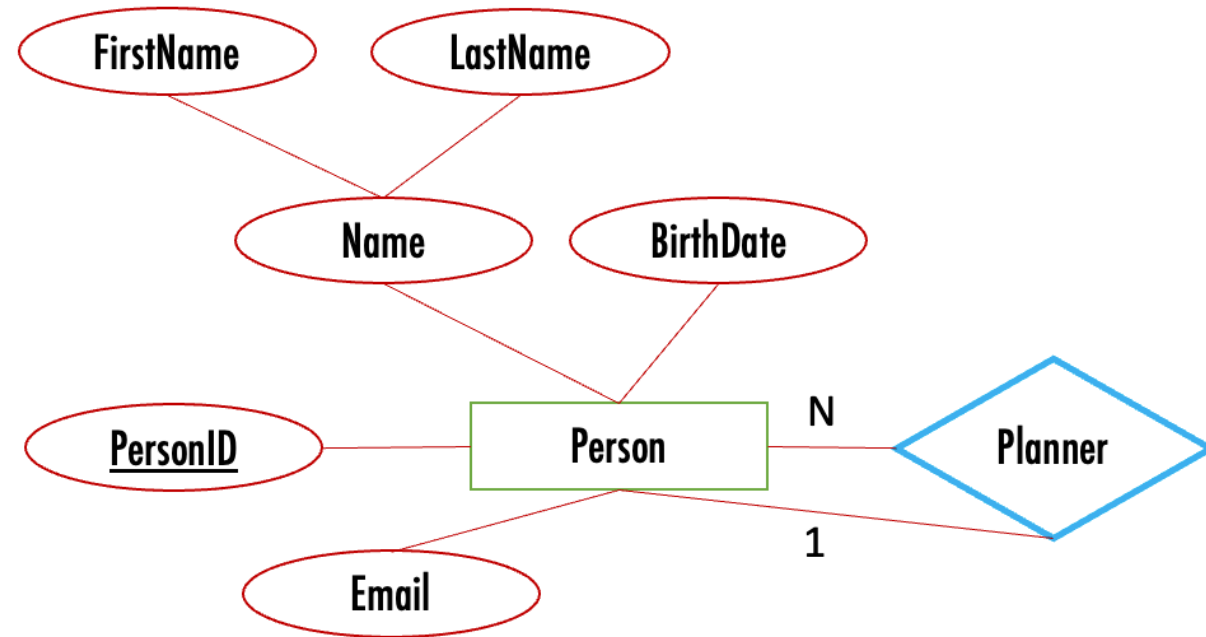
```
WHERE flight.flightNumber=flightinfo.flightNumber
```

```
GROUP BY day HAVING count(time)>1; -- What happens if we leave the GROUP BY out?
```

Running Example: birthday

- Scenario: send a message to people on their birthday, and remind their designated party planner to plan a party earlier
- Data: name (first, last), birthdate, email address, and designated party planner

Use [this file](#) to create the database.



Demonstrate writing queries - including recovering from errors!

Some queries to write (1/2)

1. Find people whose birthday is today so we can email them (combine first and last names)
2. Find people with a special birthday (e.g. 100) coming up (hint: need to calculate age, use FLOOR), order by closest birthday
3. Find people with a birthday in August, September or October so we can email their planner (hint: use IN, can also use BETWEEN)

Some queries to write (2/2)

Administrative tasks:

4. Find people with missing or invalid email addresses (hint: use IS NULL AND ... NOT LIKE)
5. Find where two (or more) people have the same email address (hint: use EXISTS)
6. Find people who are not planners for anyone else and do not have a birthday (hint: use EXISTS)

Analysis task:

7. how many people have a birthday in each month? (hint: use GROUP BY and EXTRACT(MONTH from ...))

Tips for writing SQL

- Build it up in pieces, testing each piece (* can be useful)
- Think about:
 - What information you need to see in the result (which columns ...) ... values?
Calculations? Aggregate calculations (e.g. SUM)?
 - What tables they come from, and how you need to JOIN the tables
 - What constraints to use ... and how to express them
 - WHERE if based on row
 - WHERE ... AND EXISTS – if not (or sometimes UNION, INTERSECT)

More tips

- `' '` (single quotes), not `" "` (double quote)
- DISTINCT – the whole *row* needs to be distinct, so sometimes select fewer columns
- GROUP BY ... HAVING ... - the "HAVING" goes just after the "GROUP BY ..."
- NULL is not the same as the empty string
- Too many rows in result? Check JOIN conditions
- No rows in result? Check conditions, and perhaps remove some
- Using EXISTS – make sure there is also an "AS ..." (so can refer to parent query in sub-query)

Activities

Activities – More SQL (1/4)

1. Go to learn.data.scie.nz and login if needed
2. Select MySQL (not Trino) and your own workspace (`user_[username]`)
3. If needed, create the flight DB using [this file](#)

Write queries to answer the following questions (next slides)

Activities – More SQL (2/4)

4. Retrieve the names of pilots, ordered by the earliest flight that they fly (and, if the flight time is the same, ordered by name). The desired output is on this slide (note the column name):

Hint: use ORDER BY, AS and DISTINCT

	EarlyBird 
1	Jim Belich
2	Pat Smith
3	Yi Zhang

Activities – More SQL (3/4)

5. Retrieve the flight numbers and days of flights in the first half of August 2018 (i.e. before the 16th) flown by plane models 310 or 320 (you can assume that there are no other models of the form 3n0, so can use LIKE to match against `'%3_0'`)

	flightNumber	day
1	N124	2018-08-13
2	NZ123	2018-08-10
3	NZ456	2018-08-11

Activities – More SQL (4/4)

6. Find the IDs of all pilots who fly more than one flight on a given day - try doing this first using an aggregate and GROUP BY and HAVING, and then using EXISTS.